

Autoregressive model of order 1 (AR1)

Parametrization

The autoregressive model of order 1 (AR1) for the Gaussian vector $\mathbf{x} = (x_1, \dots, x_n)$ is defined as:

$$\begin{aligned}x_1 &\sim \mathcal{N}(0, (\tau(1 - \rho^2))^{-1}) \\x_i &= \rho x_{i-1} + \epsilon_i; \quad \epsilon_i \sim \mathcal{N}(0, \tau^{-1}) \quad i = 2, \dots, n\end{aligned}$$

where

$$|\rho| < 1$$

Hyperparameters

The precision parameter κ is represented as

$$\theta_1 = \log(\kappa)$$

where κ is the *marginal* precision,

$$\kappa = \tau(1 - \rho^2).$$

The parameter ρ is represented as

$$\theta_2 = \log\left(\frac{1 + \rho}{1 - \rho}\right)$$

and the prior is defined on $\theta = (\theta_1, \theta_2)$.

Specification

The AR1 model is specified inside the `f()` function as

```
f(<whatever>, model="ar1", values=<values>, hyper = <hyper>)
```

The (optional) argument `values` is a numeric or factor vector giving the values assumed by the covariate for which we want the effect to be estimated. See the example for RW1 for an application.

Hyperparameter specification and default values

doc Auto-regressive model of order 1 (AR(1))

hyper

theta1

hyperid 14001

name log precision

short.name prec

prior loggamma

param 1 5e-05

initial 4

fixed FALSE

to.theta function(x) log(x)

from.theta function(x) exp(x)

theta2

hyperid 14002

```

    name logit lag one correlation
    short.name rho
    prior normal
    param 0 0.15
    initial 2
    fixed FALSE
    to.theta function(x) log((1 + x) / (1 - x))
    from.theta function(x) 2 * exp(x) / (1 + exp(x)) - 1
  theta3
    hyperid 14003
    name mean
    short.name mean
    prior normal
    param 0 1
    initial 0
    fixed TRUE
    to.theta function(x) x
    from.theta function(x) x

  constr FALSE

  nrow.ncol FALSE

  augmented FALSE

  aug.factor 1

  aug.constr

  n.div.by

  n.required FALSE

  set.default.values FALSE

  missing.values.warning TRUE

  pdf ar1

```

Example

In this exaple we implement an ar1 model observed with Poisson counts

```

#simulate data
n = 100
rho = 0.8
prec = 10
## note that the marginal precision would be
marg.prec = prec * (1-rho^2)

E=sample(c(5,4,10,12),size=n,replace=T)
eta = as.vector(arima.sim(list(order = c(1,0,0), ar = rho), n = n,sd=sqrt(1/prec)))
y=rpois(n,E*exp(eta))

```

```
data = list(y=y, z=1:n, E=E)

## fit the model
formula = y~f(z,model="ar1")
result = inla(formula,family="poisson", data = data, E=E)
```

Notes

A third hyperparameter θ_3 is **experimental**, and the *mean* of the AR1 process. By default this parameter is fixed to be zero.